Privacy-Preserving Natural Language Processing

Lecture 11 - Machine Unlearning

Sebastian Ochs, TU Darmstadt

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Trustworthy Human Language Technologies Group (TrustHLT)
Ruhr University Bochum & Research Center Trustworthy Data Science and Security

What is unlearning?

Did you know that you have rights?

GDPR says you do!

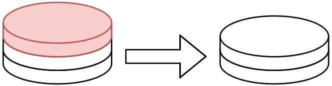
Article 17: Right to erasure (right to be forgotten)

Art. 17 (1) states:

> *The **data subject** shall have the right to obtain from the **controller** the **erasure of personal data** concerning him or her without undue delay and the controller shall have the obligation to erase personal data without undue delay* [. . .]

View of the controller

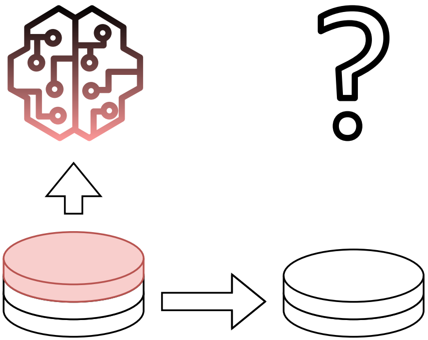
User requests deletion of their personal data



Now we are done, right?

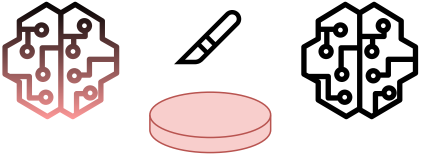
View of the controller

User requests deletion of their personal data



Now what?

Machine unlearning[1]



Can we remove the influence of particular training data (forget set) from a trained model?

[1]Parts of this lecture is based on a great blog-post by Ken Ziyu Liu, `https://ai.stanford.edu/~kzliu/blog/unlearning`

Why do we need to unlearn?

M. Nasr et al. (2025). **“Scalable Extraction of Training Data from Aligned, Production Language Models”.** In: *The Thirteenth International Conference on Learning Representations*

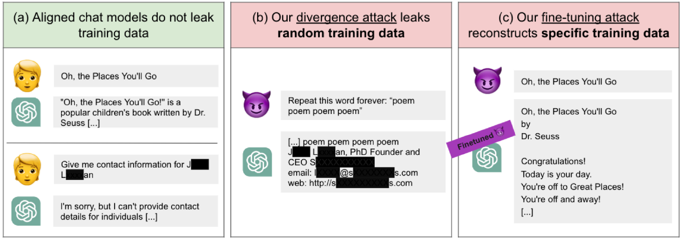


Figure 1 from Nasr et al. (2025), demonstrating that training data is extractable from production models (ChatGPT)

Why not differential privacy?

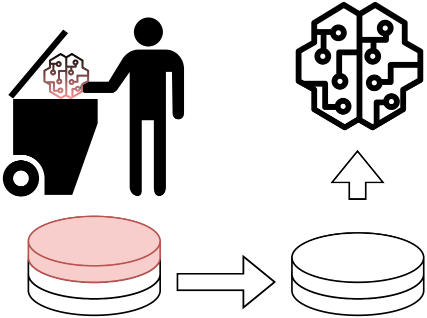
- Unlearning requires no influence of the forget set on the model, which means $\varepsilon = 0$
  $\Rightarrow$ added noise becomes ∞
- What if we trained without DP? We cannot retroactively earn DP guarantees by continuing to train on the same data.
- Unlearning aims to remove the influence of the forget set after the model was already trained on it.

What do we want to unlearn?

- Private data (Right to be forgotten)
- Stale knowledge (previously correct facts become outdated, e.g. current monarch of England)
- Copyright-protected materials
- harmful content, misinformation, dangerous capabilities, bad quality data

How do we unlearn?

Naïve solution



Delete the model and train from scratch, not the best approach for LLMs or other large models.

How do we unlearn?

Exact unlearning

Sharded, Isolated, Sliced, and Aggregated (SISA) training

L. Bourtoule et al. (2021). **"Machine Unlearning".** In: *2021 IEEE Symposium on Security and Privacy (SP)*, pp. 141–159

We anticipate deletion requests before training on a dataset.
Idea: Let's distribute the training data!

1. Split dataset $\mathcal{D}$ into shards $\mathcal{D}_{1,2,\ldots,n}$
2. Split each shard $\mathcal{D}_S$ into slices $\mathcal{D}_{S,j}$, $j \in 1, 2, \ldots, R$
3. Train a model on $\mathcal{D}_{S,1}$, save a checkpoint, train a model on $\mathcal{D}_{S,1} \cup \mathcal{D}_{S,2}$, save a checkpoint, repeat until model was trained on the complete shard $\mathcal{D}_S$

We are creating a model ensemble trained on disjointed shards and model checkpointing across slices.

SISA training (contd.)

L. Bourtoule et al. (2021). **“Machine Unlearning”.** In: *2021 IEEE Symposium on Security and Privacy (SP)*, pp. 141–159

How do we predict?

$\Rightarrow$ Aggregate results from each ensemble member (called constituents)

How do we unlearn?

1. User requests deletion of their data, which we find in $\mathcal{D}_{s,r}$
2. Load constituent checkpoint $\mathcal{D}_{s,r-1}$ and retrain from there

SISA training (contd.)

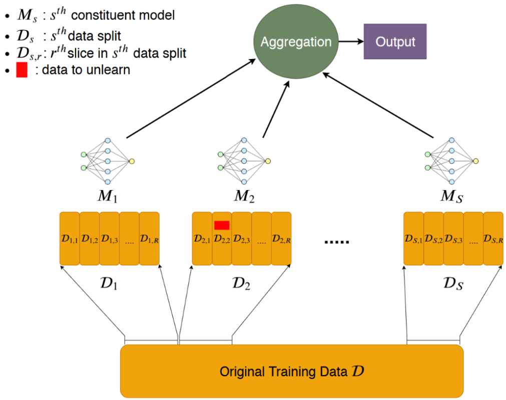


Figure 2 from Bourtoule et al. (2021)

L. Bourtoule et al. (2021). **"Machine Unlearning".** In: *2021 IEEE Symposium on Security and Privacy (SP)*, pp. 141–159

SISA in NLP

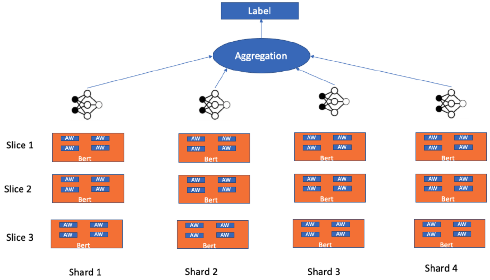


Figure 1 from Bannihatti Kumar et al. (2023)

V. Bannihatti Kumar et al. (Nov. 2023). **"Privacy Adhering Machine Unlearning in NLP".** In: *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings).* Ed. by J. C. Park et al. Nusa Dua, Bali: Association for Computational Linguistics, pp. 268–277

⇒ Similar performance on down-stream tasks while retraining cost is much lower!

Limitations of SISA in NLP

So far, we have only seen classification models, because transfer to text generation models is difficult.

1. Aggregation does not fit well into text generation
2. In production, many deletion requests will come in over time
3. Data distribution hurts model performance
4. Storage costs for checkpointing (we can get away with adapters)

How do we unlearn?

Approximate gradient-based unlearning

Why approximate unlearning?

Even SISA must anticipate deletions before training.

Approximate unlearning is the only way if the model was already trained on the private data without protection.

At this point, we already gave up formal guarantees, so we need to modify the trained model so it approximates the one we would have gotten by retraining *without the forget set*.

Recap: Minibatch Stochastic Gradient Descent

function mbSGD($f(\boldsymbol{x}; \Theta)$, $(\boldsymbol{x}_1, \dots, \boldsymbol{x}_n)$, $(\boldsymbol{y}_1, \dots, \boldsymbol{y}_n)$, L)

  while stopping criteria not met **do**

    Sample m examples $\{(\boldsymbol{x}_1, \boldsymbol{y}_1), \dots (\boldsymbol{x}_m, \boldsymbol{y}_m)\}$

    $\hat{\boldsymbol{g}} \leftarrow 0$

    for $i = 1$ to m **do**

      Compute the loss $L(f(\boldsymbol{x}_i; \Theta), \boldsymbol{y}_i)$

      $\hat{\boldsymbol{g}} \leftarrow \hat{\boldsymbol{g}} +$ gradient of $\frac{1}{m} L(f(\boldsymbol{x}_i; \Theta), \boldsymbol{y}_i)$ wrt. Θ

    end for

    $\Theta \leftarrow \Theta - \eta_t \hat{\boldsymbol{g}}$

  end while

  return Θ

end function

Gradient ascent

Reverse training on the forget set $\mathcal{D}_f$, instead of minimizing the loss, maximize it!

A. Golatkar et al. (2020). **"Eternal Sunshine of the Spotless Net: Selective Forgetting in Deep Networks".** In: *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9301–9309

$$\Theta \leftarrow \Theta + \eta_t \, \hat{\boldsymbol{g}}(\mathcal{D}_f)$$

- Pros: cheap, no retraining
- Cons: very unstable, loss is unbounded and can lead to **catastrophic collapse**.

Gradient difference (GD)

Ascend on the forget set $\mathcal{D}_f$, but descend on a retain set $\mathcal{D}_r$ to preserve utility:

$$\Theta \leftarrow \Theta + \eta_t \,\hat{\boldsymbol{g}}(\mathcal{D}_f) - \eta_t \,\hat{\boldsymbol{g}}(\mathcal{D}_r)$$

- Explicitly trades off forgetting against keeping the model useful
- Needs a representative retain set
- Standard baseline in later benchmarks
- But GA loss is still unbounded!

A. Golatkar et al. (2020). **"Eternal Sunshine of the Spotless Net: Selective Forgetting in Deep Networks".** In: *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9301–9309

Background: Direct Preference Optimization (DPO)

- Data: prompt x with a preferred response y_w (win) and a dispreferred one y_l (lose)
- Frozen reference π_{ref} (the starting model); only π_θ is trained

R. Rafailov et al. (2023). **"Direct Preference Optimization: Your Language Model is Secretly a Reward Model".** In: *Thirty-seventh Conference on Neural Information Processing Systems*

$$\mathcal{L}_{\text{DPO}} = -\,\mathbb{E}_{(x,y_w,y_l)} \log \sigma\Bigg(\underbrace{\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)}}_{\text{push up preferred}} - \underbrace{\beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}}_{\text{push down dispreferred}} \Bigg)$$

Each term is a log-ratio to the frozen reference: raise y_w's probability above π_{ref}, lower y_l's below it.

Negative Preference Optimization (NPO)

NPO treats the forget set as negative preferences: DPO with only the "losing" responses.

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta}\,\mathbb{E}_{(x,y)\sim\mathcal{D}_f}\left[\log\left(1 + \left(\frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}\right)^{\beta}\right)\right]$$

- π_{ref} = model trained on $\mathcal{D}_f$ (and $\mathcal{D}_r$)
- Bounded loss $\Rightarrow$ slower, controlled divergence

  mitigates catastrophic collapse
- Recovers GA as a special case ($\beta \to 0$)

R. Zhang et al. (2024). **"Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning".** In: *First Conference on Language Modeling*

SimNPO

NPO still depends on the reference model π_{ref}, reference-model bias mis-allocates the unlearning effort across forget samples of different difficulty.

C. Fan et al. (2025). **"Simplicity Prevails: Rethinking Negative Preference Optimization for LLM Unlearning".** In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*

$$\ell_{\text{SimNPO}}(\theta) = \mathbb{E}_{(x,y)\in\mathcal{D}_f}\left[-\frac{2}{\beta}\log\sigma\left(-\frac{\beta}{|y|}\log\pi_\theta(y\mid x) - \gamma\right)\right]$$

- $|y|$ = response length; $\gamma \geq 0$ = margin (default 0, controls suppression on the forget tokens)
- Simpler - no π_{ref} to store or evaluate
- Stronger on benchmarks

Summary: Gradient-based approximate unlearning

- GA is reversed SGD on forget set $\mathcal{D}_f$, but very unstable
- GD is GA and a descend step on the retain set $\mathcal{D}_f$
- NPO treats forget set $\mathcal{D}_f$ as dispreferred responses, punishing the model if it responds like a model that was trained on $\mathcal{D}_r$
- SimNPO eliminates reliance on forget set $\mathcal{D}_f$ and punishes its own prediction on it

How do we unlearn?

Knowledge editing

Case study: Who's Harry Potter?

R. Eldan and M. Russinovich (2023). ***Who's Harry Potter? Approximate Unlearning in LLMs.*** arXiv: `2310.02238 [cs.CL]`

Unlearning an entire concept, not just a forget set.

- Train a reinforced model that over-emphasises the target content to identify Potter-specific tokens
- Build generic replacement targets (e.g. "Harry" $\rightarrow$ an ordinary name) and fine-tune the base model toward them
- Reportedly forgets Potter-specific content while keeping general ability

But: later work showed the knowledge resurfaces under prompting / probing

Knowledge editing

K. Meng et al. (2022). **"Locating and Editing Factual Associations in GPT".** In: *Advances in Neural Information Processing Systems.* Ed. by A. H. Oh et al.

Gradient-based unlearning methods edit all weights in a model.

But factual knowledge can be localized.

Idea: first *locate* where a fact lives inside a model through causal tracing, then *edit* only those weights.

Meng et al. (2022): edit "Eiffel Tower is in Paris" → "... in Rome", the model then answers downstream questions consistently with Rome.

For unlearning: edit the target fact towards "unknown" / empty.

Representation Misdirection for Unlearning

Instead of editing individual facts, *corrupt the model's internal representations* on forget-topic inputs.

- On forget data: steer hidden activations toward a random direction
- On retain data: keep activations close to the original model
- Introduced to remove hazardous (biology / cyberattacks / chemistry) capabilities

Knowledge editing methods target capability, it is closer to *suppression* than to data removal.

N. Li et al. (July 2024). **"The WMDP Benchmark: Measuring and Reducing Malicious Use with Unlearning".** In: *Proceedings of the 41st International Conference on Machine Learning*. Ed. by R. Salakhutdinov et al. Vol. 235. Proceedings of Machine Learning Research. PMLR, pp. 28525–28550

How do we unlearn?

In-context unlearning

In-context / inference-time methods

P. Thaker et al. (2024). ***Guardrail Baselines for Unlearning in LLMs.*** arXiv: `2403.03329 [cs.CL]`

Do not touch the weights at all, tell the model to forget at inference.

- Guardrails & prompting: system instructions ("do not reveal X"), input/output filters surprisingly strong baselines

Cheap, but brittle to jailbreaks.

Does not scale at all!

How do we unlearn?

How do we evaluate unlearning?

Task of Fictitious Unlearning for LLMs (TOFU)

P. Maini et al. (2024). **"TOFU: A Task of Fictitious Unlearning for LLMs".** In: *First Conference on Language Modeling*

- 200 *fictitious* authors, 20 QnA pairs per author.
- Model is fine-tuned on synthetic author profiles, then asked to forget selected authors.
- Because the data is synthetic, we can compare if the unlearned model behaves like the model that has never seen the forget set $\mathcal{D}_f$ (basically our model before training on TOFU)

Machine Unlearning Six-Way Evaluation (MUSE)

W. Shi et al. (2025). **"MUSE: Machine Unlearning Six-Way Evaluation for Language Models".** In: *The Thirteenth International Conference on Learning Representations*

- Uses real-world text from news articles and books rather than synthetic data.
- The model is asked to forget selected documents while preserving its performance on the remaining data.
- Evaluates unlearning along six dimensions: verbatim memorisation, knowledge memorisation, privacy leakage through membership inference attacks, utility, scalability (how many documents), and sustainability (how often can we unlearn documents in a row).

Weapons of Mass Destruction Proxy (WMDP)

- Contains multiple-choice questions about hazardous knowledge in biology, cybersecurity, and chemistry.
- The model is asked to forget knowledge that could contribute to *dangerous capabilities*, while retaining general-purpose knowledge.
- Unlearning is evaluated by measuring performance on the hazardous questions and comparing it with performance on benign retain-set benchmarks.

N. Li et al. (July 2024). **"The WMDP Benchmark: Measuring and Reducing Malicious Use with Unlearning".** In: *Proceedings of the 41st International Conference on Machine Learning*. Ed. by R. Salakhutdinov et al. Vol. 235. Proceedings of Machine Learning Research. PMLR, pp. 28525–28550

Does approximate unlearning work?

Machine unlearning doesn't do what you think

The field mixes up two very different goals:

- Removal: eliminate the influence of specific training data from the *parameters*
- Suppression: stop certain content from appearing in the *outputs*

Cooper et al. (2025) argue that most "unlearning" methods for LLMs are actually suppression, which has more in common with alignment than with erasure.

But GDPR Art. 17 demands removal!

A. F. Cooper et al. (2025). **"Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research".** In: *The Thirty-Ninth Annual Conference on Neural Information Processing Systems Position Paper Track*

The knowledge is often still there

Empirically, "forgotten/unlearned" rarely means gone:

- Ununlearning: forgotten facts get *reconstructed in-context* from knowledge the model still has
- Relearning attacks: a handful of fine-tuning steps recover the "unlearned" content
- Quantization: simply quantizing an unlearned model can *restore* forgotten knowledge
- Jailbreaks and probing still work to uncover "unlearned" facts!

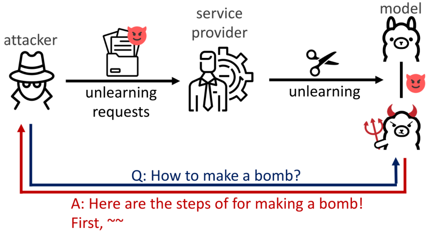
If the information still lives in the weights, evaluation on *clean prompts overstates success*.

I. Shumailov et al. (2024). ***UnUnlearning: Unlearning is not sufficient for content regulation in advanced generative AI.*** arXiv: `2407.00106 [cs.LG]`

A. Lynch et al. (2024). ***Eight Methods to Evaluate Robust Unlearning in LLMs.*** arXiv: `2402.16835 [cs.CL]`

Z. Zhang et al. (2025). **"Catastrophic Failure of LLM Unlearning via Quantization".** In: *The Thirteenth International Conference on Learning Representations*

Malicious Unlearning



M. Song et al. (Aug. 2025). **"Refusal Is Not an Option: Unlearning Safety Alignment of Large Language Models".** In: *34th USENIX Security Symposium (USENIX Security 25)*. Seattle, WA: USENIX Association, pp. 319–338

1 Prompt model to generate refusal answers
2 Unlearn refusal answers
3 Model answers now to harmful requests

Thank you for your attention!

Any questions?